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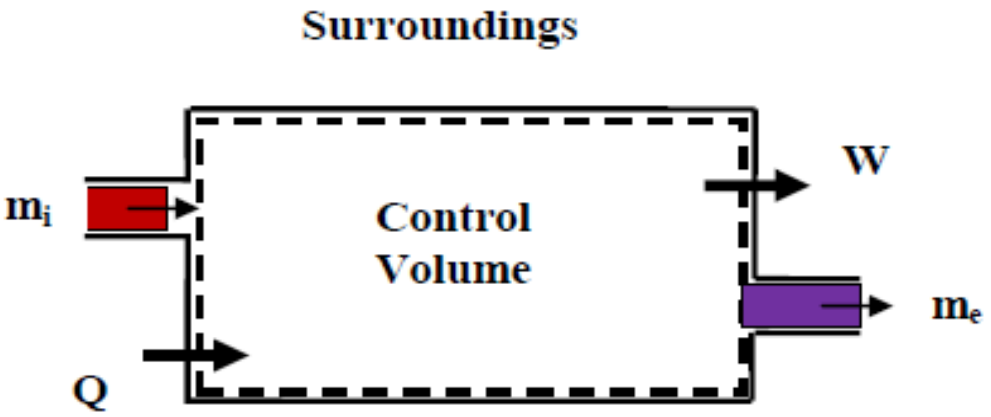
ME1100 Thermodynamics

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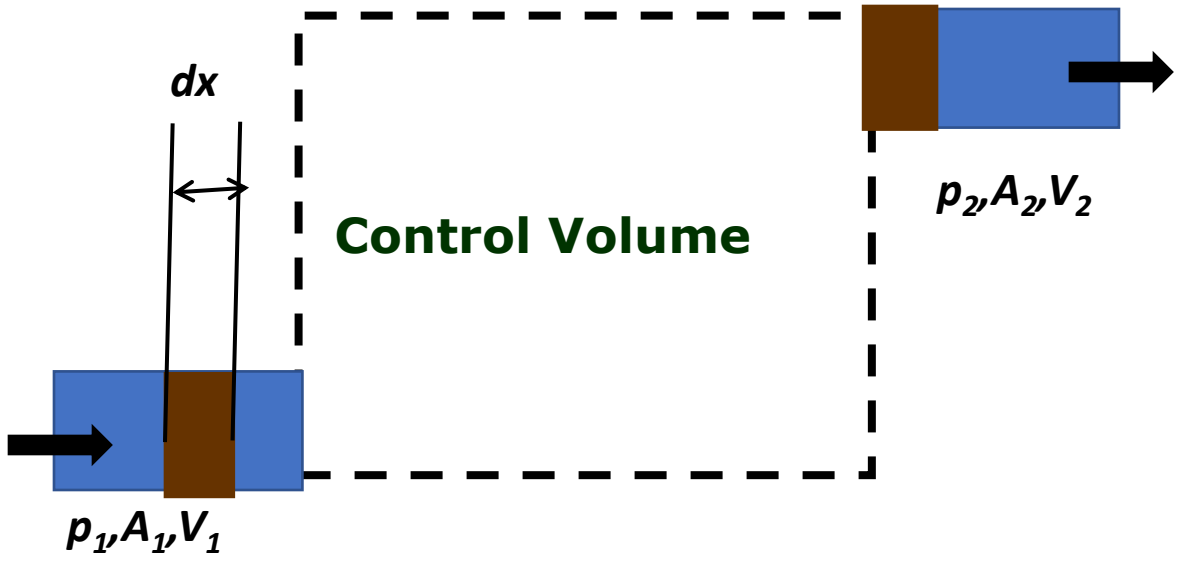
Control volume



- Control volume (CV) is defined as the fixed volume on which attention is focused.
- Apart from energy interactions it also allows in-flow and out-flow (of mass) through entry and exit ports respectively
- Practical devices that involve flow of fluid such as turbine, pump, compressor, nozzle, valve, boiler, condenser, gas cylinder, etc. can be modeled as CV.

Work interaction for CV

- For CV the work interaction does not include displacement work.
- Includes other forms of work, 'external work': shaft work, electrical work, magnetic work, etc.
- 'Internal work'



$$F_{in} = p_1 A_1 \qquad W_{flow,in} = F dx = p_1 A_1 dx$$

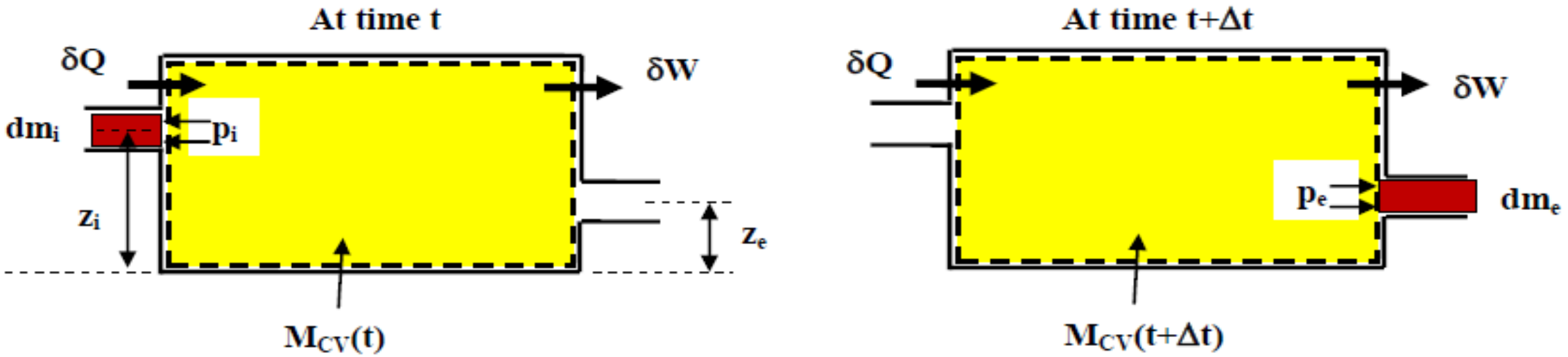
$$\frac{W_{flow,in}}{m_{in}} = \frac{p_1 A_1 dx}{\rho_1 A_1 dx} = \frac{p_1}{\rho_1} = p_1 v_1$$

$$\frac{W_{flow,out}}{m_{out}} = \frac{p_2 A_2 dx}{\rho_2 A_2 dx} = \frac{p_2}{\rho_2} = p_2 v_2$$

Flow Work (specific value): pv

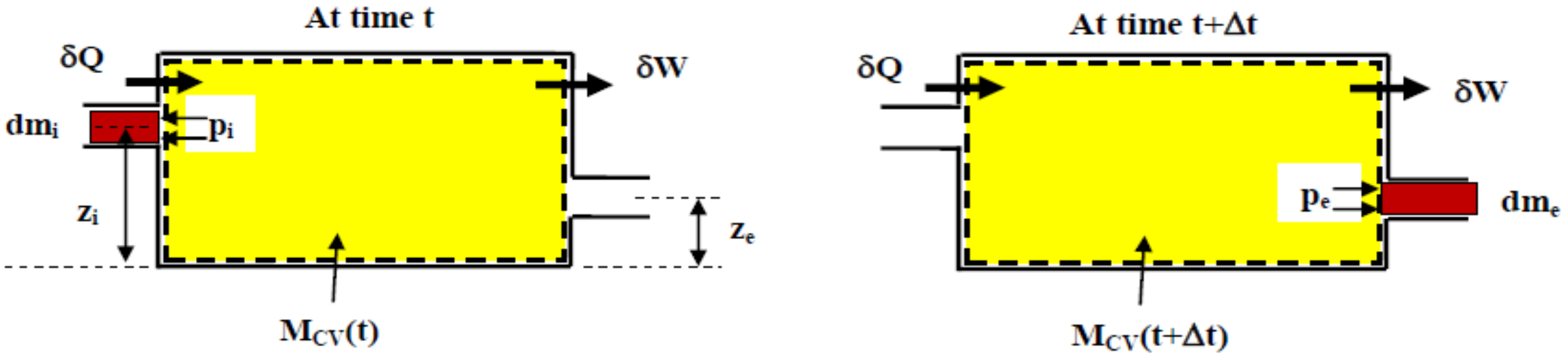
Derivation of first law for CV

- For sake of simplicity let us consider single inlet and single outlet situation.
- The basic concept involved is to write first law for a system from time ‘t’ to time ‘t+dt’ and then express it in rate form.
- For system-based approach, focus is on the same mass at the two instants of time. Then change in total energy of this system is considered that is caused by the boundary interactions of heat and work.



States of system at time instants t and (t + Δt)

Mass conservation

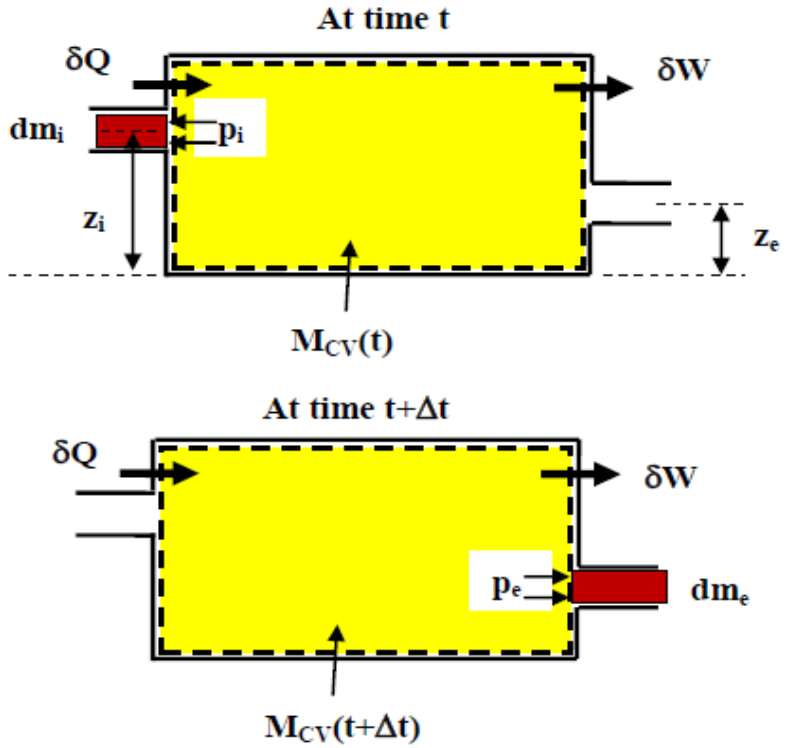


- Fluid mass ($M_{CV}(t) + dm_i$) is identified as system at time t
- Fluid mass ($M_{CV}(t+ \Delta t) + dm_e$) is identified as system at time ($t +\Delta t$)
- $M_{CV}(t)$ and $M_{CV}(t+\Delta t)$: fluid masses that occupy the control volume at time t and ($t+\Delta t$) respectively
- dm_i and dm_e are the masses that enter into and exit from the CV respectively over a time interval Δt

$$M_{CV}(t) + dm_i = M_{CV}(t + \Delta t) + dm_e$$

$$\lim_{\Delta t \to 0} \frac{M_{CV}(t + \Delta t) - M_{CV}(t)}{\Delta t} = \frac{dm_i}{dt} - \frac{dm_e}{dt}$$

$$\frac{d}{dt}(M_{CV}) = \frac{dm_i}{dt} - \frac{dm_e}{dt}$$
$$= \dot{m}_i - \dot{m}_e$$

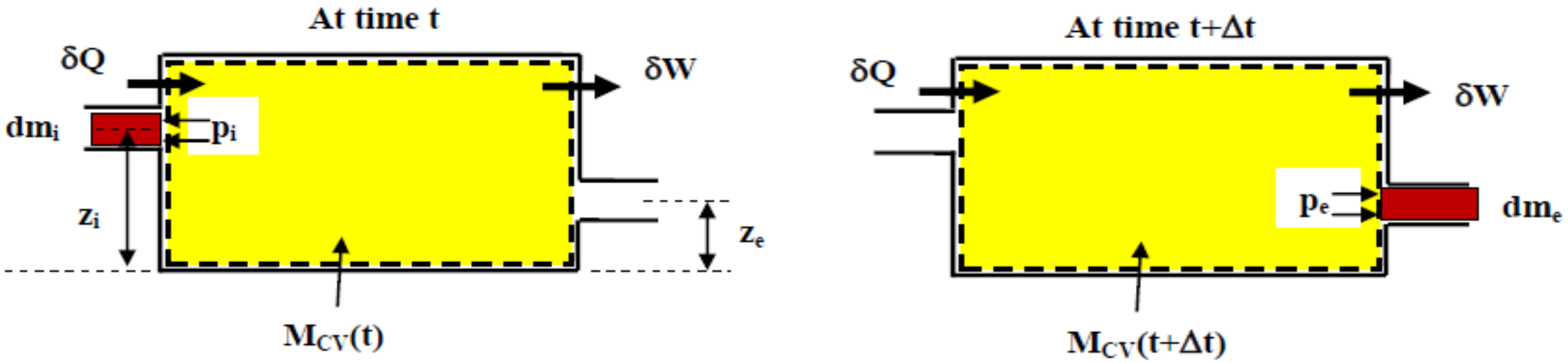


For steady state, no time dependence

$$\frac{d}{dt}(M_{CV}) = 0$$

$$\Rightarrow \frac{dm_i}{dt} - \frac{dm_e}{dt} = 0, \Rightarrow \dot{m}_i = \dot{m}_e$$

Energy Conservation

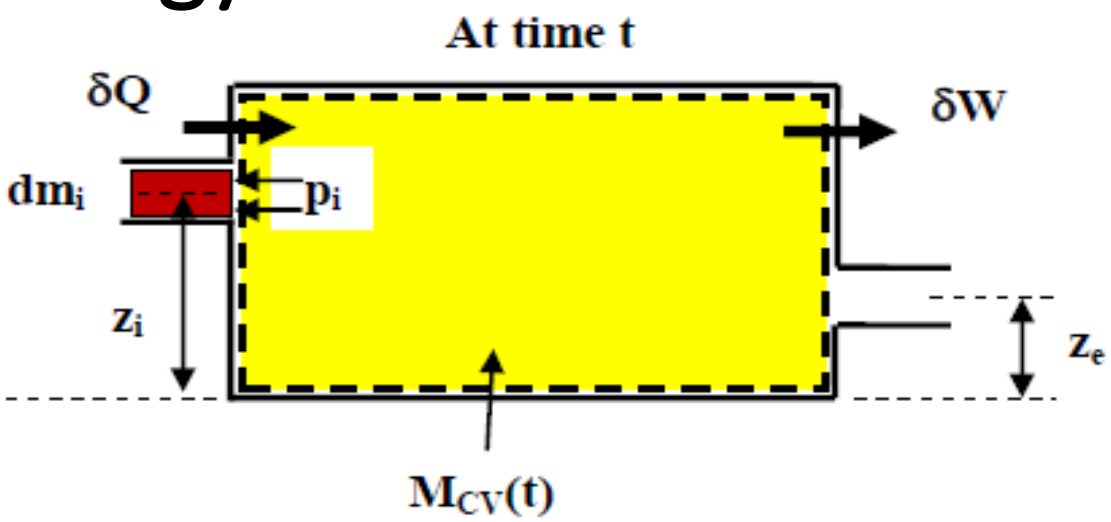


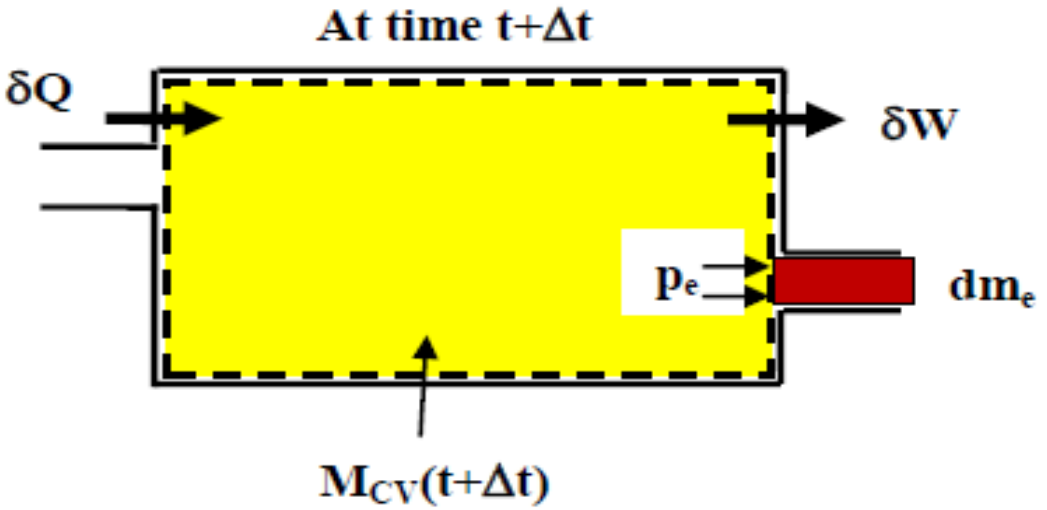
- δQ is the heat added to the system over a time interval Δt
- δW is the work done by the system over a time interval Δt

First law applied to the considered system

Heat added to the system – Work done by the system
= Change in total energy of the system

Energy conservation





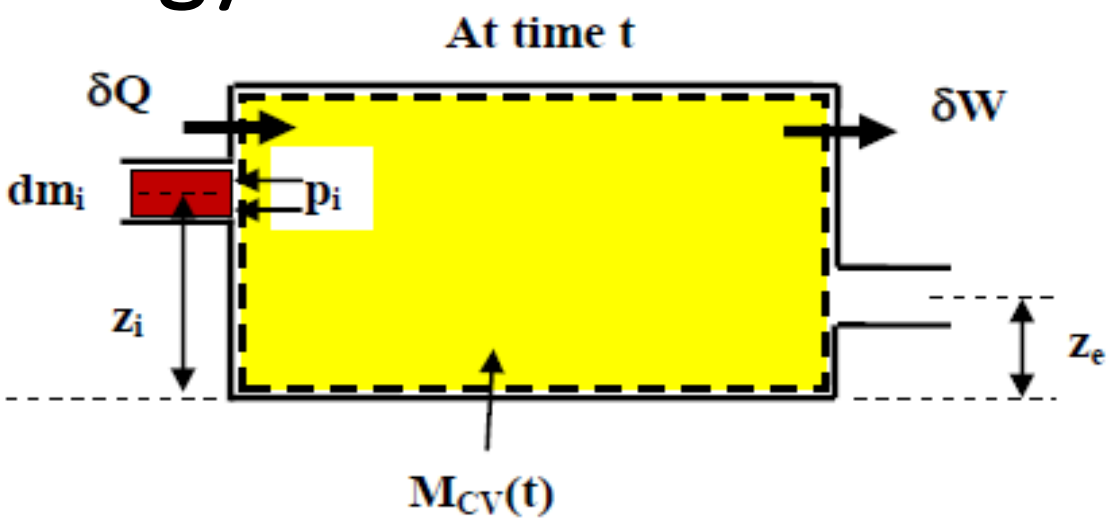
Total energy of the system at time t = $E_{CV}(t) + dm_i\left(u_i + \frac{V_i^2}{2} + gz_i\right)$

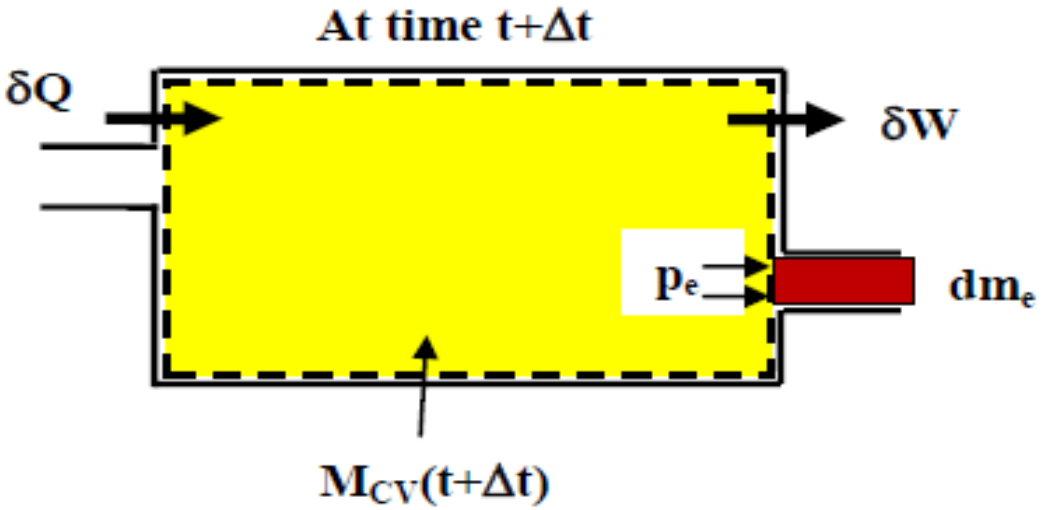
Total energy of the system at time (t+Δt) = $E_{CV}(t+\Delta t) + dm_e\left(u_e + \frac{V_e^2}{2} + gz_e\right)$

Net flow work done over a time interval Δt = $\delta W_{flow} = -dm_i\left(p_i v_i\right) + dm_e\left(p_e v_e\right)$

- Flow work at inlet is negative: surrounding pushes the fluid in.
- Flow work at exit is positive: exiting fluid pushes the surrounding outward.

Energy conservation





First law applied to the considered system

$$\delta Q - \delta W = dE$$

$$\delta Q - \left[\delta W_{ext} + dm_e\left(p_e v_e\right) - dm_i\left(p_i v_i\right)\right] = \left\{E_{CV}(t+\Delta t) + dm_e\left(u_e + \frac{V_e^2}{2} + gz_e\right)\right\} - \left\{E_{CV}(t) + dm_i\left(u_i + \frac{V_i^2}{2} + gz_i\right)\right\}$$

After rearrangement of the terms we get,

$$\delta Q - \delta W_{ext} + dm_i\left(\underbrace{p_i v_i + u_i}_{h_i} + \frac{V_i^2}{2} + gz_i\right) - dm_e\left(\underbrace{p_e v_e + u_e}_{h_e} + \frac{V_e^2}{2} + gz_e\right) = \left[E_{CV}(t+\Delta t) - E_{CV}(t)\right]$$

Diving both sides by Δt and taking limit as $\Delta t \rightarrow 0$, we get

$$\lim_{\Delta t \to 0} \frac{\left[E_{CV}(t+\Delta t) - E_{CV}(t)\right]}{\Delta t} = \frac{dE_{CV}}{dt} = Q' - W'_{ext} + \dot{m}_i \left(h_i + \frac{V_i^2}{2} + gz_i \right) - \dot{m}_e \left(h_e + \frac{V_e^2}{2} + gz_e \right)$$

This is for single inlet and single outlet ports

For multi-inlet and multi-exit ports

$$\frac{dE_{CV}}{dt} = Q' - W'_{ext} + \sum \dot{m}_i \left(h_i + \frac{V_i^2}{2} + gz_i \right) - \sum \dot{m}_e \left(h_e + \frac{V_e^2}{2} + gz_e \right)$$

Here, $E_{CV}=U_{CV}+KE_{CV}+PE_{CV}$

For a stationary control volume, KE and PE changes are negligible

$$E_{CV} = U_{CV} = (mu)_{CV} = m_{CV} u_{CV}$$

For steady state, no time dependence $\frac{d}{dt}(E_{CV}) = 0$

Steady flow devices

$$\frac{d}{dt}\left(M_{CV}\right)=0 \Rightarrow \dot{m}_i=\dot{m}_e=\dot{m}$$

$$\frac{d}{dt}\left(E_{CV}\right)=0$$

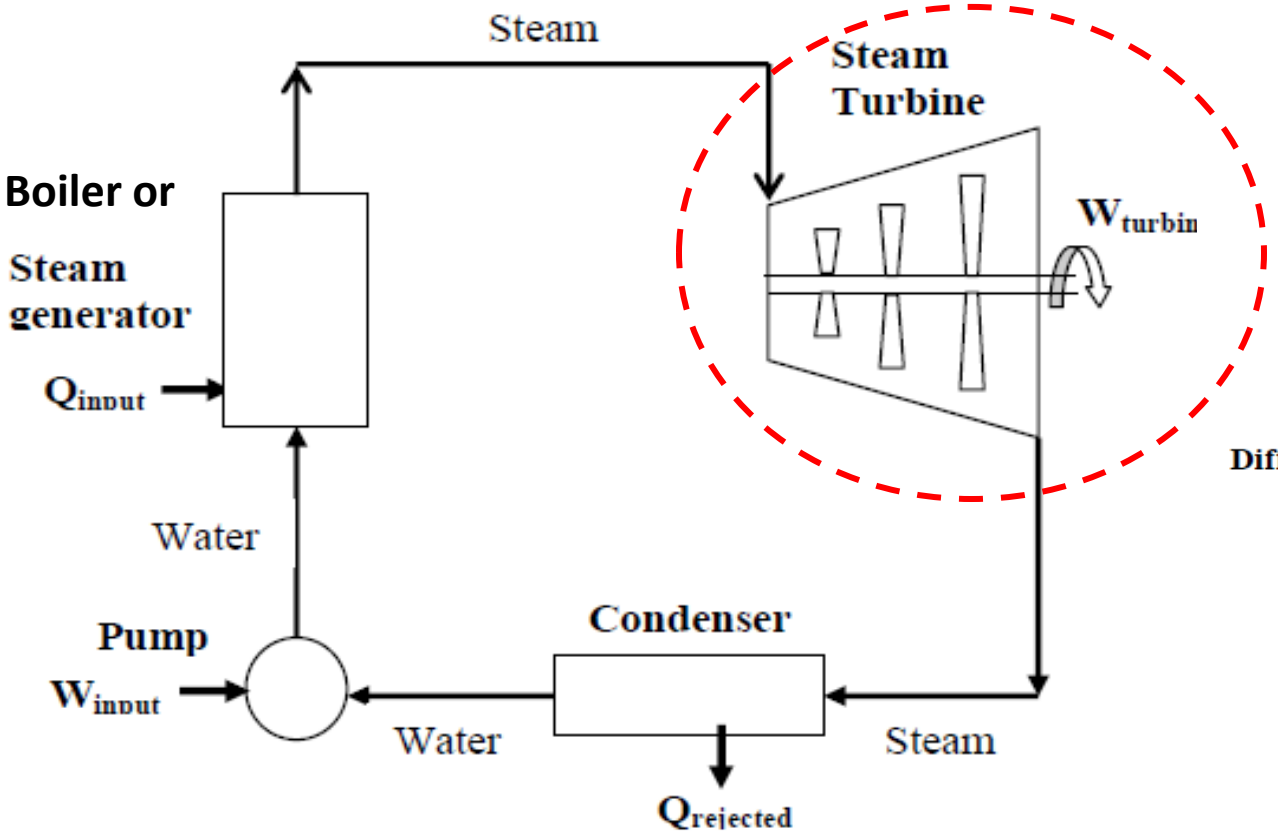
$$Q'-W'_{ext}=\dot{m}\left(h_e+\frac{V_e^2}{2}+gz_e\right)-\dot{m}\left(h_i+\frac{V_i^2}{2}+gz_i\right)$$

Steady Flow Energy Equation/
Steady state steady flow

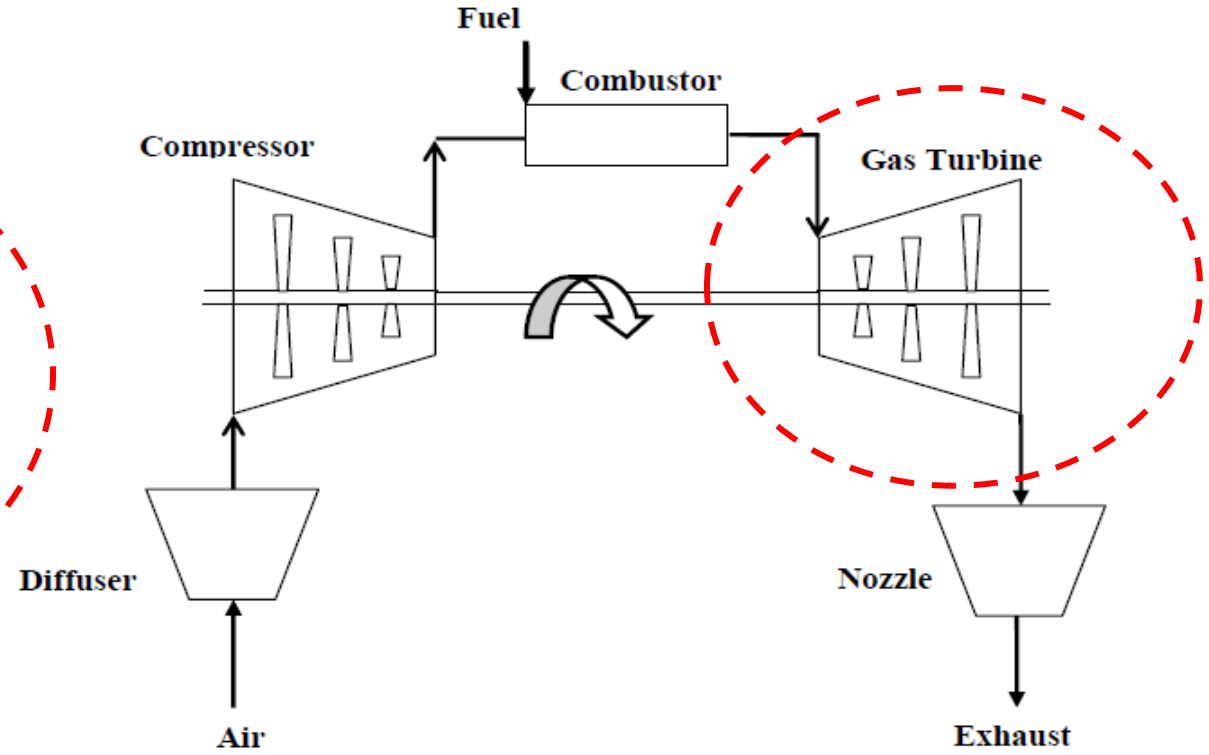
These relations are for steady flow devices with single inlet and outlet.
We may similarly write for multiple inlet/exit cases.

We will now give some examples of steady flow devices

Steady flow devices



Steam-based Power Plant

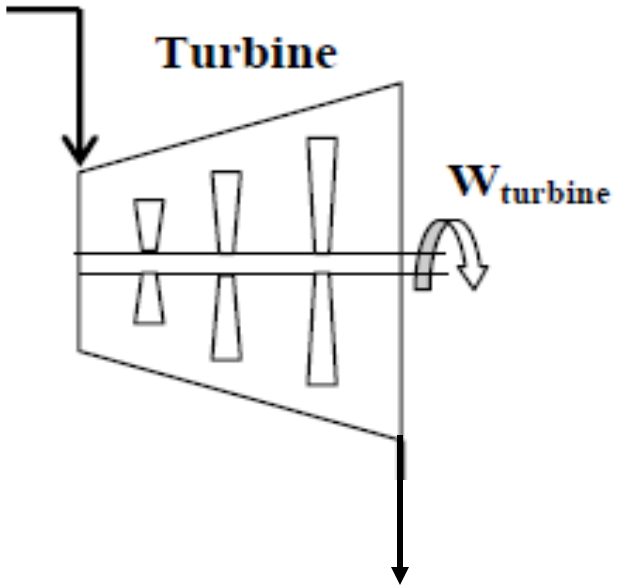


Gas-turbine based power plant

Each component can be treated as a control volume and steady-state device. Why?

Since the power plant operates in a continuous mode, except occasional shutdowns, it is meaningful to analyze individual components as steady flow devices

Steam/Gas Turbine



Assumptions:

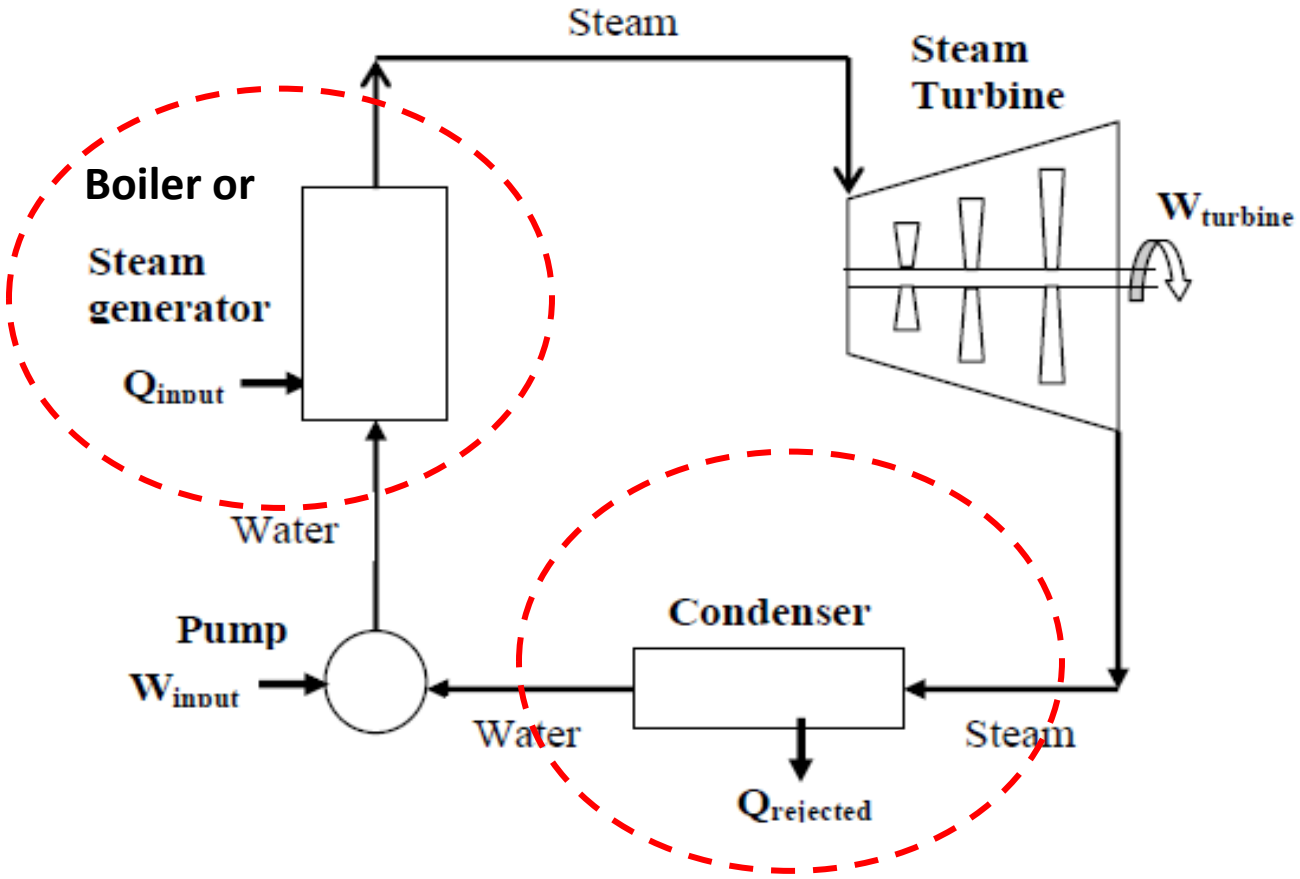
1. Heat transfer is usually neglected (adiabatic condition)
2. Changes in potential energy may be less important
3. Many times changes in kinetic energy may be neglected

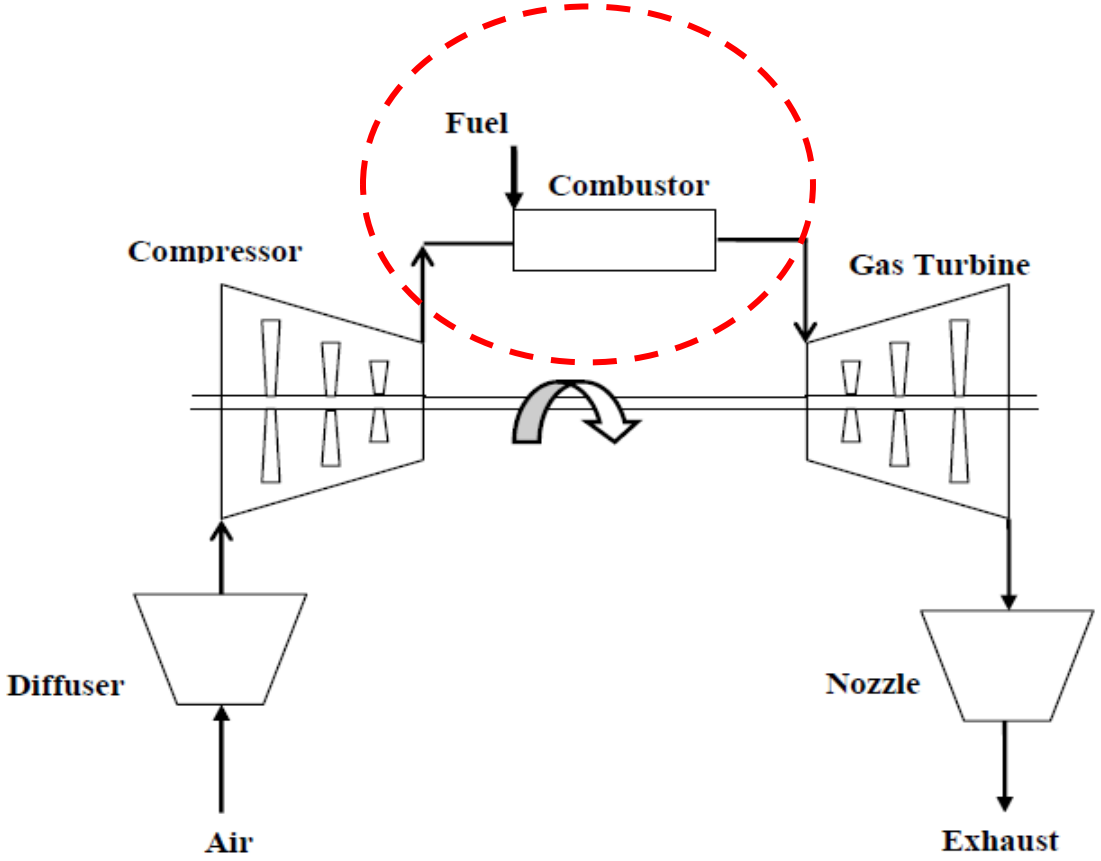
Applying Steady Flow Energy Equation

$$Q' - W'_{ext} = \dot{m}\left(h_e + \frac{V_e^2}{2} + gz_e \right) - \dot{m}\left(h_i + \frac{V_i^2}{2} + gz_i \right)$$

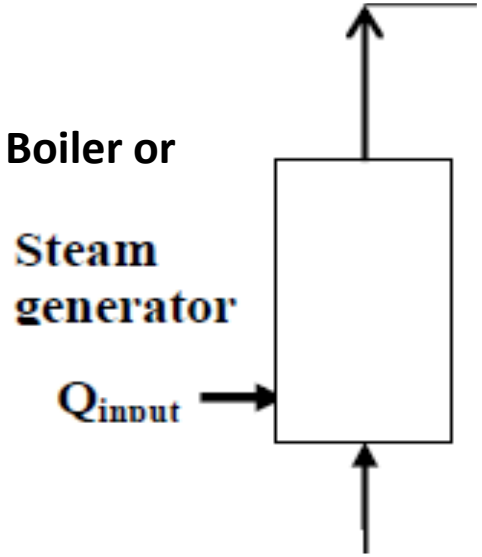
Power **produced** by the turbine: $W'_{ext} = \dot{m}\left(h_i - h_e \right)$

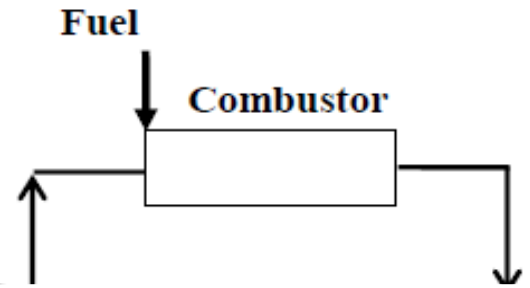
Steady flow devices

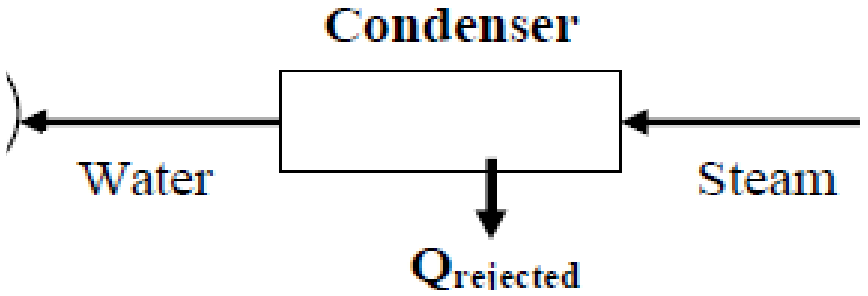




Heat Exchanger







Assumption:

No external work is done in the device and KE as well as PE changes are negligible

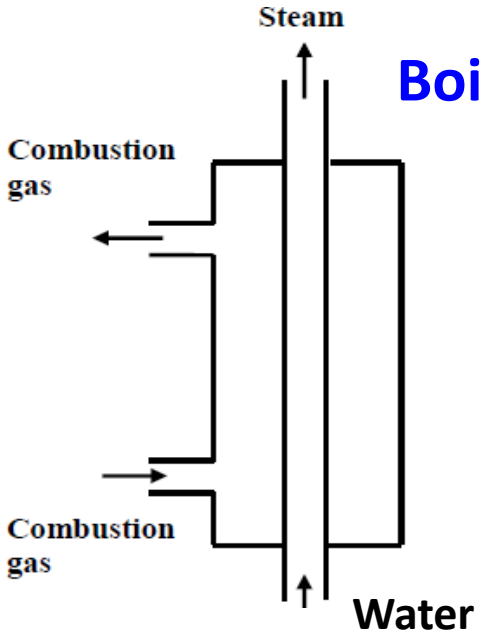
Applying Steady Flow Energy Equation

$$Q' - W'_{ext} = \dot{m}\left(h_e + \frac{V_e^2}{2} + gz_e\right) - \dot{m}\left(h_i + \frac{V_i^2}{2} + gz_i\right)$$

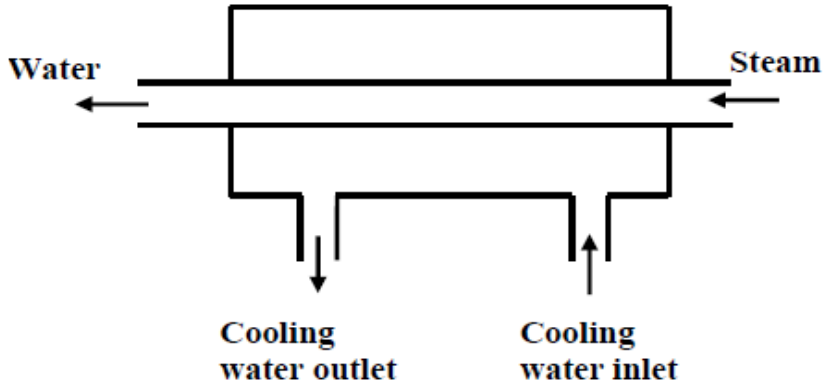
Rate of heat transfer:

$$Q' = \dot{m}\left(h_e - h_i\right)$$

Heat Exchanger



Boiler



Condenser

Assumption:

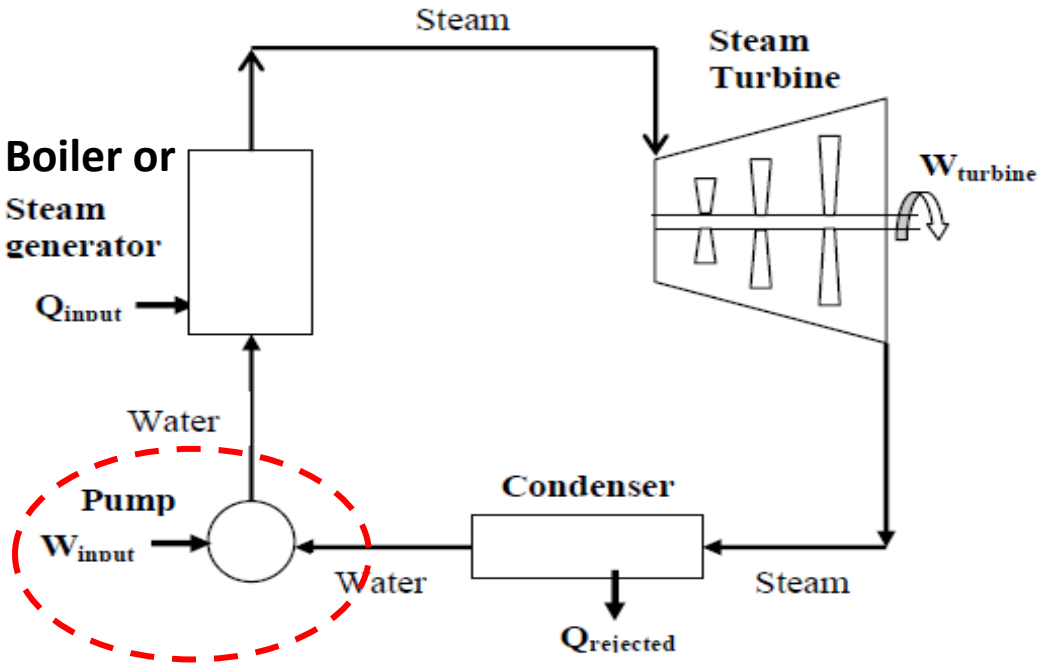
1. No external work is done in the device and KE as well as PE changes are negligible
2. Multiple inlet and outlet (considering both fluids)
3. No exchange of heat between the control volume and the surrounding

Applying Steady Flow Energy Equation

$$Q' - W'_{ext} = \sum \dot{m}_e \left(h_e + \frac{V_e^2}{2} + gz_e \right) - \sum \dot{m}_i \left(h_i + \frac{V_i^2}{2} + gz_i \right)$$

$$\sum \dot{m}_e (h_e) - \sum \dot{m}_i (h_i) = 0$$

Water pump



Assumption:

1. No heat transfer ➔ adiabatic condition
2. Liquid is almost incompressible and change in specific volume is negligible
3. Change in temperature may be negligible
4. Often, a change in KE and PE may be neglected.

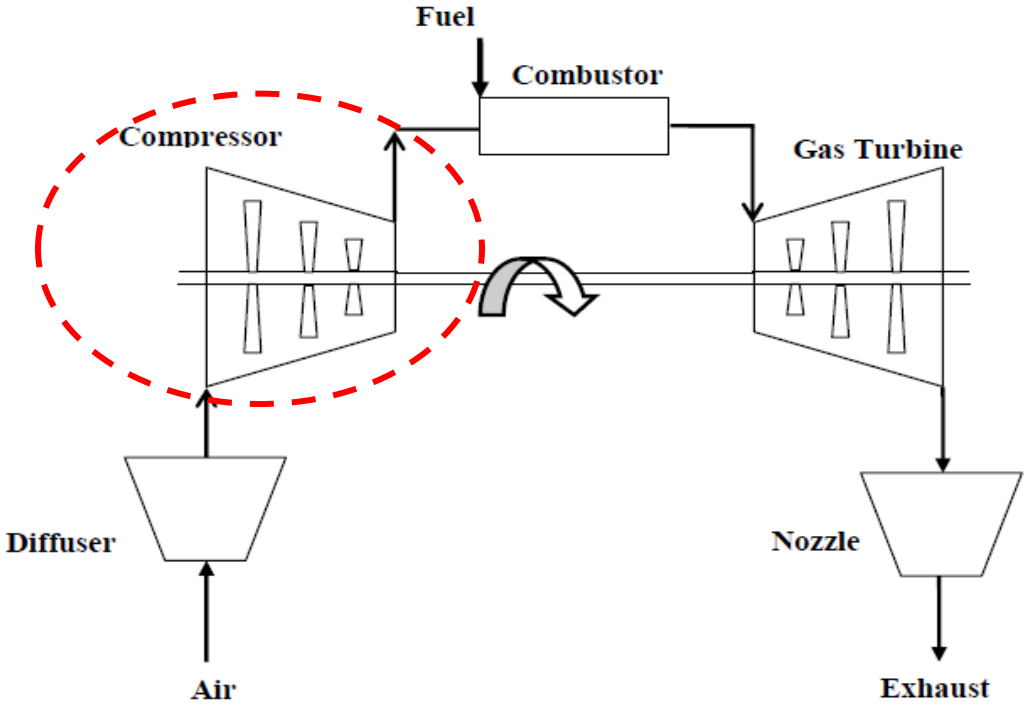
Applying Steady Flow Energy Equation

$$Q' - W'_{ext} = \dot{m}\left(h_e + \frac{V_e^2}{2} + gz_e\right) - \dot{m}\left(h_i + \frac{V_i^2}{2} + gz_i\right)$$

Power **required** by the pump:

$$W'_p = -W'_{ext} = \dot{m}\left(h_e - h_i\right) \Rightarrow W'_p = \dot{m}\left(v_f\right)_{T_f}\left(p_e - p_i\right)$$

Compressor



Assumptions:

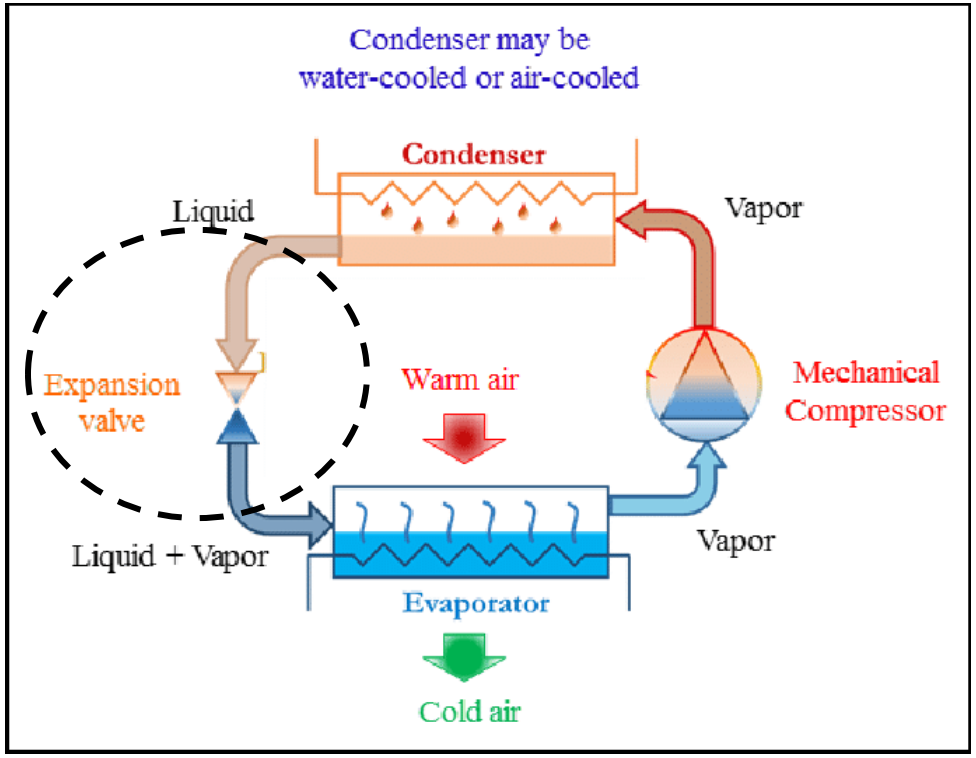
1. Heat transfer is usually neglected (adiabatic condition)
2. Changes in potential energy may be less important
3. Many times changes in kinetic energy may be neglected

Applying Steady Flow Energy Equation

$$Q' - W'_{ext} = \dot{m}\left(h_e + \frac{V_e^2}{2} + gz_e\right) - \dot{m}\left(h_i + \frac{V_i^2}{2} + gz_i\right)$$

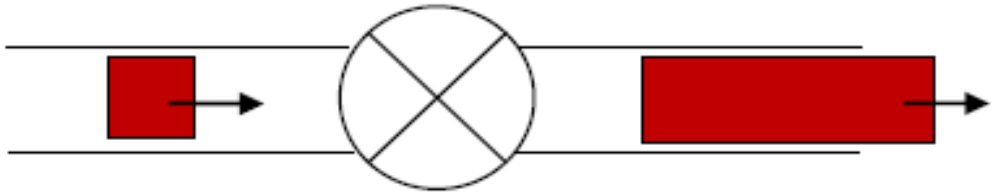
Power **needed** by the compressor: $$W_c' = -W'_{ext} = \dot{m}(h_e - h_i)$$

Vapour compression Refrigeration Cycle



https://www.researchgate.net/figure/Schematic-diagram-of-a-typical-vapor-compression-refrigeration-cycle-17_fig1_326272160

Expansion Valve



Assumptions

1. This device is used to control flow rates in pipelines and other flow equipment.
2. There are no heat and work interactions and KE as well as PE changes are assumed to be negligible

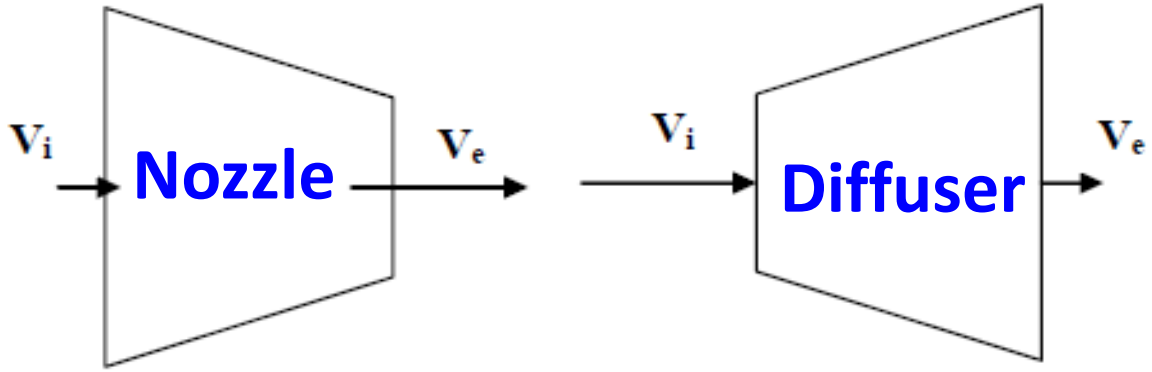
Applying Steady Flow Energy Equation

$$Q' - W'_{ext} = \dot{m}\left(h_e + \frac{V_e^2}{2} + gz_e\right) - \dot{m}\left(h_i + \frac{V_i^2}{2} + gz_i\right)$$

$$h_e = h_i$$ ⟵ Isenthalpic process

Flow Nozzles and Diffusers

Conservation of mass: $\dot{m} = \rho_i A_i V_i = \rho_e A_e V_e$



Applying Steady Flow Energy Equation

$$Q' - W'_{ext} = \dot{m}\left(h_e + \frac{V_e^2}{2} + gz_e\right) - \dot{m}\left(h_i + \frac{V_i^2}{2} + gz_i\right)$$

Assumptions

1. There are no heat and work interactions
2. PE changes are assumed to be negligible
3. KE changes **cannot** be ignored
4. Many times,

a) In nozzle, inlet velocity << outlet velocity

b) In diffuser, inlet velocity >> outlet velocity

$$h_e + \frac{V_e^2}{2} = h_i + \frac{V_i^2}{2}$$

$$h_{0e} = h_{0i}$$

h_0=stagnation or total enthalpy`